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"A Solution in Search of a Problem" is a Low-Rates Phenomenon

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IoT's Newsletter cross-posted a post from **The Diff**

IoT Mote
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I'd like to say that my article, Remaking the Nokia 6110 on 22nm is a new general purpose technology, because it converges the potential for solar power management integration on a mobile form factor. Seeing is believing. When Steve Jobs visited Xerox Parc and saw the GUI demo by Doug Englebardt, he saw a market for one in every household. He didn't say, this is only a Business to Business tool. Today, you can buy a phone with maybe a 24 hour battery life. But Dennard's scaling has allowed more computation in the same space, which also means that one can use the same amount of computation in a smaller space, with less power. And if you jsut want to make a phone call or use a word processor, you could, on solar power. But omnibus operating systems like Android and iOS do not allow this.

"A Solution in Search of a Problem" is a Low-Rates Phenomenon

Plus! Money and Equilibria; eSports; The Price of Liquidity; Plausible Deniability; The Illiquidity Premium, Continued; Diff Jobs

BYRNE HOBART
DEC 12

CROSS-POST

This is the weekly free edition of *The Diff*! Last week, paying subscribers read about a potato company whose shares are up 34% this year ($), whether or

not to worry about $80 trillion worth of opaque swaps ($), and how Grab used a taxi service to build a developing market super-app ($).

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Welcome to the free weekly edition of *The Diff*! This newsletter goes out to 48,000+ readers.In this issue:

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"A Solution in Search of a Problem" is a Low-Rates Phenomenon

In the beginning of any technology's existence, you'll hear two closely-related critiques of new technologies: that they're just a toy, or that they're a solution in search of a problem. It happens often enough that tech boosters just see the critique as part of the lifecycle of any new product: first, it's impossible; next, it's a frivolous toy that only appeals to reach hobbyists or nerds; then it's a bubble that, while promising, suffers from wildly inflated expectations; soon enough, it's become nearly ubiquitous and suddenly the product is a human right and you're a monster if you'd trade better earnings this quarter for depriving someone of it.

Regardless, we shouldn't leave this as a morality tale about the inevitable

wrongheadedness of techno-pessimists. Yes, the car, the plane, and the personal computer were all dismissed as pointless toys. So was the pet rock. More to the point, so were products that ultimately turned out to be dead ends, like the Palm Pilot or the Aibo.

The toy-to-transformative question is partly determined by the details of technologies, which can sometimes be unpredictable. Was it obvious, in advance, that GPUs that were basically designed to draw lots of triangles as efficiently as possible would turn out to be exceptionally good tools for large language models, image generation, and other applications of AI? It wasn't what the industry was expecting. But even if a smart futurist fifteen years ago speculated on it, they might have concluded that the volume of computation required by GPT-3 and its ilk was so absurdly high that it constituted a *reductio ad absurdum* demonstrating that this was not a path towards getting computers to create human-like work.[1]

One intuition pump for determining the difference between toys and tech is to consider whether the technology does something that would be useful if everyone had it, but is relatively useless at small scale. The Internet was a cool demo of networking computers together for a while, but since computers were expensive products designed for the use cases of the big institutions that could afford them, hooking them together didn't present much obvious utility—okay, so General Motors' payroll computer can talk to American Airlines' reservation computer which can talk to a computer at Oak Ridge running physics simulations, but what are these computers going to say to one another?

This is not a computer-specific phenomenon: the history of timekeeping devices in Western Europe is a few centuries of increasingly accurate clocks being incorporated into astronomical devices, and then a generation or two in which timekeeping would take off in some society and quickly come to dominate it. A factory can't easily tell people when their shift starts, or tell suppliers when their order will be delivered, if no one is used to thinking in terms of specific rather than approximate times. But once time is widely deployed, it's unavoidable; groups have to coordinate by time, and everyone who isn't a complete loner then has to build at least part of their schedule around when other people are expecting them.

The underlying characteristics of a new technology can determine potential,

and in wide-ranging ways—there's a sense in which the ancestor to Twitter's, Instagram's, Snapchat's, and TikTok's voracious demand for constant attention from users who can access them from anywhere is all descended from the very trivial and definitely-a-toy Tamagotchi. Which, if nothing else, showed that people can be highly responsive to low-bandwidth positive feedback from electronic devices, which they'll keep on their person at all times. But there's a feature of the outside world that also has a major impact: solutions-in-search-of-a-problem and toys-in-search-of-real-world-use are both less costly and more valuable in a world of lower real interest rates.

There are actually two broad reasons for this, one on the supply-of-capital side and one on the demand-for-innovation side.

On the supply side, low rates just mean that time matters less, and specifically that events in the far future are, for discounted cash flow analysis purposes, happening closer to the present. The usual way to look at this is to take a predicted stream of cash flows over time, whether it's the money you get from extrapolating a SaaS company's unit economics or just looking at the payoff of a treasury bond. But it also applies to *uncertain* events. A financial product with a 5% annual chance of a $1bn payoff is worth more when rates are low, and less when rates are high.

The other side of this equation is to look at the demand side. High real interest rates are a loud and clear message from the world's investing opportunities to the world's current and potential savers, stating that there are numerous opportunities for highly probable profits. When a country is in the midst of catch-up growth, there are many kinds of investments that have a clear payoff just by extrapolating GDP growth and looking at how spending changes as people get richer—as GDP per capita goes from $2k to $20k, penetration rate of refrigeration, indoor plumbing, cars, and appliances as GDP per capita goes from something close to zero to something in the 50-100% range. Meanwhile, growth in these countries justifies lots of infrastructure: new housing; new roads; ports, railroads, and airports; electricity generation; etc.

The flipside of this is that low rates imply that the easy investments have all been made, and that future investments just aren't that compelling. They might be less rewarding, since there's more competition, or they might be riskier, since they aren't following an existing template. Either way, they're not

attractive, and savers' money is more tempted by low-risk sovereign debt than by growth investments.

But this mostly means that the economy is asking technology to find some sort of money sink that can profitably absorb the vast amount of savings that a rich economy can produce. General-purpose technologies tend to be big capital sinks because they create demand for so much associated infrastructure. Only a small fraction of the spending created by the internal combustion engine actually went to the manufacture and maintenance of internal combustion engines; the economy also had to produce cars, roads to drive them on, oil to fuel them, pipelines and refineries for the oil, steel and other metals for all of the above, and countless other expenditures—even housing, the classic unaffected-by-technology investment class, was in fact affected by this one, both because of higher demand for new dwellings in centers of the industry (like Detroit) and its adjacent supply chains (Houston), but because cars changed the most efficient layout for cities.

One interesting trait of toys-turned-tools is that they seem more likely than average to become such general-purpose technologies. General-purpose technologies are fairly rare, and very important—many comprehensive lists of them will start with things like fire, writing, and the use of tools to improve other tools. A toy that finds another use case is more likely to be a general-purpose technology, in part because of the "Pirahã Are Off By One " phenomenon: most inventions are useless, some have an extremely specific use case, but if you discover that they have more than one use case, it's likely that this generalization will continue. And this is especially true from the standpoint of someone who hears about a new tool: it's going to be more viral if a) it spreads by being useful, and b) it spreads in as many different groups as possible, i.e. it has lots of use cases. I hear about AI from my marketing friends ("instant long-tail copy on any topic!") from my finance friends ("*way* better transcript searches") and of course from my tech friends ("Code generating AI is going to make me 200% more productive from now until the world ends in a few years.")

This model of rates-affected innovation is useful in a few senses. First, it implies that once the world has a mature financial system with lots of savers—a phenomenon that's quite modern!—there may be long cycles of higher and lower innovation. A general-purpose technology in the midst of global deployment will push real rates up, which sucks capital away from more

theoretical applications. It also helps to explain some of crypto's volatility in a narrow way: crypto is doubly sensitive to real rates. First, since tokens typically don't generate immediate income (and since many of the ones that do are funding speculation), crypto token valuation tends to be rates-sensitive. But the more abstract crypto bull thessi—all the people who switched from laser eyes on Twitter to "I'm more interested in blockchain technology" on LinkedIn—is *also* a rates bet. It's a bet that crypto will find a killer app and become a general-purpose technology with widespread implications. This is another way to frame the paradox that crypto used to read as a risk-off asset that investors would rush to when they were paranoid and pessimistic about the state of the world, since it's hard to debase through inflation and hard for governments to seize. But, especially in recent years, crypto has performed more like a risk-on asset, moving in line with the Nasdaq.

This model isn't going to catch every rate hiking and rate cutting cycle, and it won't be a good way to time when the CPI rolls over and inflationary trends reverse. But it *is* a good way to think about the bigger, longer-term cycles. One- and five-year bull markets and bear markets can have many causes, but generation-long swings towards or away from wealth creation tend to depend on either entirely new technologies or new models for deploying them. In aggregate terms, the tech wealth creation cycle was not as big a deal for the world as the one induced by globalization (search is great, and social media is fun, but neither saved as many people from poverty and starvation as the death of Mao). The low-rate environment of the 2010s may have been the ideal incubator for general-purpose, growth-driving technologies that will power the next few decades. So, keep an eye on the toy products that could only get funded because rates were so low for so long; the market was asking for transformative technology, and might have gotten it.

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Money and Equilibria

One fun crypto parlor game is to predict when Tether will collapse. It's widely agreed that Tether has been unbacked for at least some of its existence, and any currency that tries to maintain a stable peg but doesn't always have the resources to do so will eventually come undone. But there are degrees of unbacking: if everyone owns Tether because 1 USDT is backed by 1 USD, then it can't collapse, but if every USDT is backed by, say, 95 cents instead, then it's possible to have a run on the bank, since every redemption at 100 cents on the dollar makes it less backed. But that run can only happen if people are worried about such a run.

In one model, there are really two constituencies: people who want to speculate in crypto, and who treat Tether as convenient for this, and people who want to preserve dollar-denominated wealth but not interact with the US banking system. For those people, Tether can be compared to asset classes like art and real estate, where the round-trip cost to convert the savings vehicle back into dollars has a substantial cost and uncertainty. So for them, unbacked Tether may still be better than the alternatives. If the size of that population's Tether savings is larger than any hole in Tether's balance sheet, it can run at negative equity indefinitely (not are these holders less likely to redeem for dollars, but they're less likely to lend out their Tether, so it's hard

for someone to launch a campaign to bust the peg).

One possible catalyst for weakening Tether is to convince a large population of users to transfer their funds into some similar but better-backed asset, and this is something Coinbase is trying to do by offering no-fee conversion between Tether and their own USDC. One reading of this is that it's a bare-knuckled way to hurt a competing stablecoin, but another possibility is that Coinbase has a vested interest in crypto stability—not just because of their dependence on trading commissions, but because the brunt of regulations will be borne by the regulated players, so a big collapse in a poorly-regulated corner of crypto can have government-induced contagion on the rest of the system. Ensuring that more legitimate Tether holders have fewer legitimate reasons to hold Tether reduces the potential damage if Tether does indeed come unpegged.

Disclosure: Long BTC, and long Coinbase bonds.

eSports

The eSports business is losing both revenue and hype. As it turns out, 2020-1 was a perfect situation for the industry:

- There was a boom in home entertainment in general and video games in particular.
- Crypto trading created more sponsorship opportunities; as it turns out, there are domains besides League of Legends where having quick reaction time and a willingness to stare at screens all day can pay off.
- SPACs created exit opportunities, and that naturally inflated the value of private companies.
- A general increase in social media use helped eSports on the marketing side as well as monetization.

Now that all of that is reversing, the industry suddenly looks like it has a cost structure approaching that of professional sports, but without the decades of carefully cultivating monopolistic access to specific events that have loyal, multi-generational fanbases. Meanwhile, the core fundamental of a growing media business—viewership—is moving in the wrong direction, too. This kind of reversal can quickly crush valuations; FaZe Holdings was worth $1.8bn in August and has a market cap of $160m today.

For an earlier *Diff* look at FaZe—written when short sellers were paying 678% annualized to borrow the stock—see here ($).

The Price of Liquidity

Treasury bonds are getting less liquid, but this is mostly because rates are finally moving again ($, *FT*). Liquidity providers get paid partly for the risk of taking a big position in exactly the wrong direction. As Agustin Lebron points out, every outcome to a limit order is a disappointment: if you place a limit order to buy and you never get filled, prices are moving up and you should have paid more to go along for the ride; if you do get your order filled, prices are moving down and you're losing money as they do. And when rates are more volatile while there's extensive debate about the tightening cycle, a recession, and the real trend in long-term inflation, that adverse selection gets more common and the cost of liquidity rises to compensate.

Plausible Deniability

This *Politico* piece is a good roundup of the increasing alignment between hacker gangs and governments, especially in Russia and Ukraine. Crime produces deadweight loss, and hosting criminals in one country so long as they only commit crimes in *other* countries tends to irritate the rest of the world. But it also gives the host country an arsenal that can be valuable in times of stress. (The hackers are fighting for their freedom, but in the very literal sense that if they don't hack the government's preferred targets they're going to spend an unpleasant time in jail.) *The Diff* wrote about this phenomenon a few years ago ($), with a comparison to France's policy with respect to French mercenaries in Africa—tolerating them in order to have plausibly deniable military influence in the region, but reining them in when their actions conflicted with policy goals.

The Illiquidity Premium, Continued

A few weeks after Blackstone gated withdrawals from its real estate fund, the company is slowing down fundraising for a private equity fund targeting wealthy individual investors ($, *FT*). The economics of gating withdrawals are fascinating: they're a way for a fund to control its liquidity, by ensuring that it doesn't need to launch a fire sale at an inopportune time to get investors their cash.[2] But exercising this option, even when it's right there in the contract the investor signed, is costly in terms of future fundraising. The converse of this is

also true: one reason private equity dry powder wasn't heavily deployed in late 2008 and early 2009 was that limited partners were financially stressed, and making a capital call, even a contractually allowable one, would have upset them. Anything that is *almost* equivalent to having cash on hand will be completely different from having cash on hand only at the moment at which you'd really prefer to have cash on hand.

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1 It's a bit like Greek astronomers considering heliocentrism but rejecting it because the stars didn't measurably change positions during different seasons, implying either that the earth was stationary or that the stars were some comically absurd distance from the earth. They were right about the conditional! It just turned out that nature is more comfortable with absurd distances than humans are.

2 A common problem! In many rogue trading cases, most of the losses come from rapidly liquidating the rogue trader's position after it's discovered, not from the trades themselves. A rogue trader is very sensitive to market impact, since it's one more way they can get caught. But a company with a rogue trader problem is much more sensitive to the perception that these trades were tacitly allowed, and getting them off the books as soon as possible is a good way to allay that.

CROSS-POST

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